

MovieLens ml-latest-small

610 users x 9,724 rated movies
100,836 ratings, 98.3% sparse

TMDB plot overviews

9,603 of 9,724 movies (98.8%)
joined via links.csv tmdbId

Preprocessing

user-item rating matrix + text corpus

Per-user 60 / 20 / 20 split (seed = student ID)

train: fit validation: select hyperparameters test: report ONCE

SVD

scipy svds, $k = 10$
damped item bias

I-AutoRec

PyTorch autoencoder
masked MSE loss

Content

TF-IDF over plot text
+ cosine similarity

Nested Hybrid (deployed model)

per-user z-score blend, $\alpha_2 = 0.6$

Inner Hybrid

SVD $\oplus$ AutoRec, $\alpha_1 = 0.4$

Content

second signal

Also compared

Hybrid (weighted)
SVD $\oplus$ Content, $\alpha = 0.5$

Hybrid (switching)
SVD if warm, else Content

Most-Popular baseline
no personalization

Evaluation on the held-out test split

Accuracy RMSE / MAE Ranking Precision@K, Recall@K, NDCG@K Beyond-accuracy coverage, novelty, serendipity
Responsible AI popularity bias, cold-start reach Robustness the whole comparison repeated across 4 splits